Notes: Dessein and Santos (JPE, 2006)

Adaptive Organizations

Contents

1 Big picture

- **Question.** How should organizations choose the division of labor when workers have local information and specialized tasks must be coordinated with each other?
- **Setting.** The paper builds a team-theoretic model of an organization that must perform n interdependent tasks. Each task needs to be adapted to local circumstances, but adaptation by one worker can create coordination problems for other workers.
- **Core puzzle.** Classic theories often predict that better communication technology should increase specialization because specialized workers can coordinate more easily. But many modern workplace practices move in the opposite direction: broader jobs, more teamwork, more employee autonomy, and more horizontal communication.
- **Main theoretical idea.** Organizations choose not only *how specialized* to be, but also *how adaptive* to be. This extra choice changes the specialization–communication trade-off.
- **Two organizational modes.**
  - **Rigid / specialized organization:** workers follow a preagreed action plan, local information is mostly ignored, and coordination is achieved *ex ante*. This allows high specialization.
  - **Adaptive organization:** workers use local information, actions are adjusted to circumstances, and coordination must be achieved *ex post* through communication and broader task assignments.
- **Main contribution.** The paper endogenizes the organization's choice of adaptiveness, task bundling, and communication. It shows that broad job design, employee flexibility, and intensive communication can be complementary organizational practices.
- **Why it matters.** The model rationalizes why improvements in communication and information technology can be associated with **less** specialization, not more. Better communication can make adaptation attractive, and adaptation itself creates a need for broader jobs.
- **Main results.**
  - More environmental uncertainty or a greater importance of adaptation leads to broader jobs and more employee flexibility.
  - Higher Smithian returns to specialization lead to narrower jobs and less flexibility.
  - More task interdependence does *not* always imply broader jobs. When coordination becomes very important, the organization may instead suppress flexibility and coordinate by rigid rules.
  - Better communication technology can reduce specialization when it pushes the organization from a rigid mode into an adaptive mode.
  - With endogenous communication, broad jobs, employee flexibility, and communication intensity move together.
  - Organizational design is often convex: organizations tend to cluster into either rigid-specialized-low-communication systems or adaptive-broad-high-communication systems.
- **Economic takeaway.** The division of labor cannot be understood without asking whether the organization wants to use workers' local knowledge. Specialization is not limited only by coordination costs; it is limited by the organization's desired degree of adaptiveness.

2 Data and measurement (key objects)

2.1 This is a theory paper: what the key objects are

- There is **no empirical dataset**. The important objects are model primitives: tasks, local information, primary actions, complementary actions, task bundling, communication quality, adaptation costs, coordination costs, and specialization costs.
- The paper studies three linked organizational choices:
  - how many tasks each worker performs,
  - how much flexibility workers receive to adapt to local information,
  - how much the organization relies on horizontal communication.
- The model is deliberately simple: all employees share the same organizational objective. The paper abstracts from incentives and politics in order to isolate the adaptation–coordination–specialization mechanism.

2.2 Tasks, local information, and actions

- The organization must carry out n tasks.
- Each task i has a local state or local information variable v_i.
  - v_i is observed only by the worker responsible for task i.
  - v_i has mean $\hat{v}_i$ and common variance σ_v^2.
  - The realizations of v_i are independent across tasks.
- Each task has one **primary action** a_{ii}.
  - Perfect adaptation requires $a_{ii} = v_i$.
  - If a_{ii} is close to v_i, task i is well adapted to its local environment.
- Each task also has complementary actions a_{ij} for $j \neq i$.
  - These actions are needed to coordinate task i with the primary actions of other tasks.
  - Perfect coordination of task i with task j requires the relevant complementary action to track the other task's primary action.
- **Interpretation.** In product design or software development, one team may adapt its module to a local problem. Other teams then need to know that change in order to keep their own modules compatible.

2.3 Task specialization and task bundling

- Each task is assigned to exactly one employee, but an employee can be assigned several tasks.
- $T(i)$ denotes the set of tasks bundled with task i.

- The organization is restricted to symmetric task assignments: every worker performs the same number t of tasks.
- The feasible set of task bundles is

$$J = \{t \in \mathbb{N} : n/t \in \mathbb{N}\}.$$

- A larger t means **more task bundling** and therefore **less specialization**.
- A smaller t means **more specialization**. The extreme cases are:
  - $t = 1$: each employee performs one task, full specialization;
  - $t = n$: one employee performs all tasks, full task bundling.
- Task bundling is costly in the Smithian specialization sense. The labor cost of carrying out task i is $h(t, \alpha)$, increasing in t, where α measures the returns to specialization.

2.4 Communication quality

- If two tasks are assigned to different employees, the relevant local information must be communicated.
- Communication is imperfect. With probability p, the receiving worker understands the message; with probability $1 - p$, the message is pure noise.
- p is the quality of the communication channel.
  - Low p: difficult horizontal communication.
  - High p: easier and more reliable communication.
- If tasks are bundled under the same worker, the worker observes the relevant local information directly, so there is no communication failure inside the bundle.
- In the first part of the model, p is exogenous. Later, the organization can choose p by paying a communication cost.

2.5 Timing

1. **Organizational design stage.** The organization chooses the number of tasks per worker, t.
2. **Information stage.** Local conditions v_i are realized. The worker responsible for task i observes v_i.
3. **Communication stage.** Workers attempt to communicate local information. Each cross-worker communication succeeds with probability p.
4. **Action stage.** Workers choose primary and complementary actions subject to their information constraints.

Interpretation. The organization chooses the architecture first. Local conditions then arrive, communication takes place, and workers choose actions.

2.6 Main parameters and their economic roles

Object	Meaning	Higher value means
ϕ	Importance of adaptation	Local fit matters more
β	Importance of coordination	Task interdependence is stronger
σ_v^2	Variance of local information	Environment is more uncertain
α	Returns to specialization	Task bundling is more costly
p	Communication quality	Horizontal communication is more reliable
δ	Communication-cost parameter	Communication is more expensive
t	Tasks per worker	Broader jobs / less specialization

3 Models and empirical specifications (all equations + interpretation)

General note. This is a theory paper. The core "specifications" are the payoff function, the optimal action rules, the employee-flexibility measure, and the comparative statics for task bundling and communication.

3.1 Two-task example: adaptation and coordination losses

For two tasks, adaptation and coordination losses are

$$\phi\big[(a_{11}-v_1)^2+(a_{22}-v_2)^2\big]+\beta\big[(a_{12}-a_{22})^2+(a_{21}-a_{11})^2\big].$$

- ϕ weights adaptation losses.
- β weights coordination losses.
- a_{11} and a_{22} are primary actions.
- a_{12} and a_{21} are complementary actions.

Interpretation. A worker wants to adapt his own primary action to local information, but other workers must coordinate complementary actions with it. Adaptation creates useful local fit but can create cross-task misalignment.

3.2 General organizational profit and task-level costs

Organizational profits are negative expected production costs:

$$-\sum_{i=1}^{n} C^i(a_{1i},a_{2i},\ldots,a_{ni},t \mid v_i),$$

where

$$C^i(a_{1i},\ldots,a_{ni},t \mid v_i)=\phi(a_{ii}-v_i)^2+\sum_{j\notin T(i)}\beta(a_{ji}-a_{ii})^2+h(t,\alpha).$$

- The first term is the adaptation loss for task i.
- The second term is the coordination loss created when task i is not bundled with task j.
- The third term is the labor cost from assigning t tasks to a worker.

Interpretation. Task bundling eliminates communication failure inside the bundle. If task i and task j are controlled by the same worker, coordination between them is automatic in this baseline model.

3.3 Specialization costs and difference notation

The paper uses the difference notation

$$\Delta f(\bar{t}, t) = f(\bar{t}) - f(t), \qquad \Delta_x f(\bar{t}, t) = f_x(\bar{t}) - f_x(t).$$

Smithian returns to specialization imply that for $\bar{t} > t$,

$$\Delta h(\bar{t}, t) \geq 0, \qquad \Delta_\alpha h(\bar{t}, t) \geq 0.$$

Interpretation. Moving from t to $\bar{t}$ broadens jobs. Broader jobs are more costly, and this cost increase is larger when returns to specialization are stronger.

3.4 Optimal primary and complementary actions

Given a task-bundling level t and communication quality p, optimal primary actions satisfy

$$a_{ii}(t) = \hat{v}_i + \left[\frac{\phi}{\phi + \beta(n-t)(1-p)}\right](v_i - \hat{v}_i).$$

Complementary actions satisfy

$$a_{ji}(t) = \begin{cases} a_{ii}(t), & \text{when task } j \text{ learns } v_i, \\ \hat{v}_i, & \text{when task } j \text{ does not learn } v_i. \end{cases}$$

Interpretation.

- The primary action is a weighted average of the ex ante plan $\hat{v}_i$ and the local information v_i.
- The weight on local information is

$$\frac{\phi}{\phi + \beta(n-t)(1-p)}.$$

- This weight is higher when adaptation is more important, when communication is better, or when more tasks are bundled.
- This weight is lower when coordination is more important or when there are more unbundled tasks to coordinate with.

3.5 Employee flexibility

Employee flexibility is defined as the covariance between the primary action and the local information:

$$\sigma_{av}(t) = \mathrm{Cov}(a_{ii}(t), v_i) = \left[\frac{\phi}{\phi + \beta(n-t)(1-p)}\right] \sigma_v^2.$$

Interpretation.

- $\sigma_{av}(t)$ measures how much workers actually use local information.
- $\sigma_{av}(t) = 0$ would mean workers do not respond to local conditions.
- $\sigma_{av}(t) = \sigma_v^2$ means full adaptation to local conditions.
- $\sigma_{av}(t)$ increases with task bundling t because bundled tasks are automatically coordinated.
- $\sigma_{av}(t) = \sigma_v^2$ if $t = n$ or $p = 1$; in either case the adaptation–coordination trade-off disappears.

3.6 Expected profit and optimal task bundling

The expected profit from task bundling level t is

$$P(t) = -\mathbb{E}\left[\sum_i \min_{\bar{a}^i} C^i(\bar{a}^i, t \mid v_i)\right] = n\phi\big[\sigma_{av}(t) - \sigma_v^2\big] - nh(t, \alpha),$$

where $\bar{a}^i = (a_{1i}, a_{2i}, \ldots, a_{ni})$.

The optimal level of task bundling is

$$t^* = \arg\max_{t \in J} P(t).$$

Interpretation.

- The first part captures adaptation and coordination performance.
- The second part captures the cost of giving workers broader jobs.
- The organization trades off higher flexibility and better ex post coordination against lost specialization.

3.7 Specialization versus adaptation

From the profit expression, the returns to broader jobs increase with the importance and variability of local information:

$$\Delta_\theta P(\bar{t}, t) \geq 0 \qquad \text{for } \theta \in \{\phi, \sigma_v^2, -\alpha\}.$$

Main result.

- Task specialization decreases when ϕ rises.
- Task specialization decreases when σ_v^2 rises.

- Task specialization increases when α rises.

Interpretation. When local adaptation is more valuable, the organization gives workers broader tasks and more flexibility. When specialization gains are larger, the organization sacrifices adaptation and keeps jobs narrow.

3.8 Task interdependence and the nonmonotonicity of specialization

The effect of the coordination parameter β is ambiguous. The paper shows that

$$\Delta_\beta P(\bar{t}, t) \leq 0 \quad \text{if and only if} \quad \phi \leq \beta(1-p)(n-t)^{1/2}(n-\bar{t})^{1/2}.$$

Interpretation.

- When adaptation is very important, more task interdependence tends to favor more task bundling because coordination failures become costly.
- When adaptation is not very important, more task interdependence can make the organization more rigid. The organization reduces flexibility, making task bundling less useful.
- For very high task interdependence, the organization may choose ex ante coordination: workers follow the plan, do not adapt much, and can remain highly specialized.

The limiting result is that, for a cutoff $\hat{\phi}$,

$$\lim_{\beta \to \infty} t^* = \begin{cases} n, & \text{if } \phi > \hat{\phi}, \\ 1, & \text{if } \phi < \hat{\phi}. \end{cases}$$

Takeaway. More coordination need does not mechanically imply less specialization. The organization can coordinate either by bundling tasks ex post or by suppressing adaptation ex ante.

3.9 Communication quality and the nonmonotonicity of specialization

The effect of communication quality p is also ambiguous. The paper shows that

$$\Delta_p P(\bar{t}, t) \geq 0 \quad \text{if and only if} \quad \phi \leq \beta(1-p)(n-t)^{1/2}(n-\bar{t})^{1/2}.$$

Interpretation.

- A direct effect of better communication is to make specialization easier: workers can coordinate while remaining specialized.
- But better communication also makes adaptation more attractive: if workers can communicate local knowledge, the organization wants them to use it.
- Greater adaptation then increases the need for task bundling.

Result. When communication is initially poor, improvements in communication can lead to **less specialization**. Only when communication is already very good does further improvement tend to increase specialization in the standard way.

3.10 Endogenous communication

The organization can choose communication quality p at cost $g(t, p, \delta)$, where

$$g_p \geq 0, \qquad g(t, 0, \delta) = g_p(t, 0, \delta) = 0, \qquad \lim_{p \to 1} g_p(t, p, \delta) = \infty.$$

The communication-cost parameter satisfies

$$g_\delta > 0, \qquad g_{p\delta} > 0.$$

Profits become

$$P(t, p) = n\phi\big[\sigma_{av}(t, p) - \sigma_v^2\big] - nh(t, \alpha) - g(t, p, \delta),$$

and the organization solves

$$(t^*, p^*) = \underset{t \in J,\, p \in [0,1]}{\arg\max}\ P(t, p).$$

Interpretation. Communication is now an organizational design variable. Workers can spend time in meetings, calls, email, job rotation, or other mechanisms that improve mutual understanding, but these activities are costly.

3.11 Communication technologies

The paper discusses three ways to model communication costs.

1. **Task-to-person communication:**

   $$g(t, p, \delta) = n\left(\frac{n}{t} - 1\right) m(p, \delta).$$

   Each task's information must be communicated to other employees.

2. **Task-to-task communication:**

   $$g(t, p, \delta) = n(n - t)m(p, \delta).$$

   Each unbundled task pair requires direct communication.

3. **Person-to-person communication:**

   $$g(t, p, \delta) = \frac{1}{2}\frac{n}{t}\left(\frac{n}{t} - 1\right) m(p, \delta).$$

   Communication costs are tied to pairs of employees rather than pairs of tasks.

Interpretation. Under many natural communication technologies, broader jobs reduce the number of communication channels that must be maintained.

3.12 Complementarity between broad jobs and communication

Define communication cost per unbundled task pair as

$$\tilde{g}(t, p, \delta) = \left[\frac{2}{n(n-t)}\right] g(t, p, \delta).$$

Assumption 1 requires

$$\Delta_p \tilde{g}(\bar{t}, t) = \tilde{g}_p(\bar{t}, p, \delta) - \tilde{g}_p(t, p, \delta) \leq 0.$$

Interpretation. The marginal cost of improving communication per unbundled task pair is weakly lower when workers have broader jobs.

Given this assumption and quasiconcavity in p, the paper obtains:

$$t^* \text{ and } p^* \text{ increase in } \phi \text{ and } \sigma_v^2, \qquad t^* \text{ and } p^* \text{ decrease in } \alpha.$$

Interpretation. When the environment is more variable or local adaptation matters more, the organization adopts a cluster of practices: broader jobs, more flexibility, and more communication.

3.13 Supermodularity and clustered organizational practices

The paper expresses the complementarity by considering

$$\pi(t, \hat{p}) = \max_{p \geq \hat{p}} P(t, p).$$

For relevant parameters $\theta \in \{\sigma_v^2, \phi, -\alpha\}$, the optimized profit function is supermodular in t, $\hat{p}$, and θ.

Interpretation. Broader jobs raise the returns to better communication, and better communication raises the returns to broader jobs. This generates clusters of organizational practices rather than smooth, one-at-a-time changes.

3.14 Appendix extension: technological trade-offs

The appendix adds direct technological conflict between adaptation and coordination. The generalized cost is

$$C^i = \phi(a_{ii} - v_i)^2 + \beta \sum_{j \neq i} (a_{ji} - a_{ii})^2 + \lambda \sum_{j \neq i} (a_{ij} - a_{ii})^2 + h(t, \alpha).$$

- $\lambda = 0$ gives the baseline model.
- $\lambda \to \infty$ gives a one-action-per-task model in which primary and complementary actions must coincide.

In this extension,

$$\sigma_{av}(t) = \mu_\phi(t)\sigma_v^2,$$

where $\mu_\phi(t)$ is increasing in task bundling t.

Interpretation. The main insight survives: broader jobs increase employee flexibility. The extension also shows that, when technological trade-offs are strong, too much emphasis on adaptation can eventually make coordination irrelevant and restore specialization.

4 Identification and instruments (what makes the estimates credible)

4.1 What standard approaches miss

- Standard specialization theories emphasize the productivity gains from narrower jobs and the communication costs of coordinating specialized workers.
- In those theories, better communication usually means that specialized workers can coordinate more cheaply, so specialization should rise.
- This logic misses the endogenous choice of adaptiveness. Once better communication lets organizations use local information, workers become more flexible, and flexibility itself creates new coordination demands.
- The paper's key move is to make adaptation an organizational choice rather than a fixed requirement.

4.2 What identifies the mechanism here

The mechanism is pinned down by three linked ingredients:

1. **Local information:** only the worker assigned to a task observes that task's local state.
2. **Imperfect communication:** specialized workers cannot always transmit local information reliably.
3. **Costly task bundling:** broader jobs solve some coordination problems but sacrifice specialization gains.

Interpretation. None of these elements alone generates the paper's main results. The results come from their interaction: local information makes adaptation valuable, imperfect communication makes adaptation costly under specialization, and task bundling creates an alternative coordination technology.

4.3 Why the comparative statics are sharp

- A rise in ϕ or σ_v^2 directly raises the value of using local information. This makes flexibility more valuable and increases the return to task bundling.
- A rise in α raises the cost of broad jobs. This pushes the organization toward specialization and reduced flexibility.
- A rise in β has two opposing effects. It increases the cost of miscoordination, but it also makes the organization want to reduce flexibility so that coordination is easier.
- A rise in p also has two opposing effects. It makes specialization easier, but it also makes adaptation more attractive.

Interpretation. The unambiguous results come from parameters that shift the value or cost of adaptation directly. The ambiguous results come from parameters that change both the coordination technology and the organization's desired flexibility.

4.4 Why convexity matters

- Broader jobs improve coordination.
- Better coordination lets the organization give workers more flexibility.
- More flexibility raises the value of further improvements in coordination.

Interpretation. This feedback creates convexity in organizational design. A moderately adaptive organization may be worse than either a rigid specialized organization or a highly adaptive organization.

4.5 Internal robustness

- The results do not rely on communication being exogenous; the complementarity becomes stronger when communication is chosen endogenously.
- The results do not rely entirely on the baseline assumption that bundling removes the adaptation–coordination trade-off. The appendix allows technological trade-offs and preserves the core mechanism that broader jobs increase flexibility.
- The model does not require incentive conflicts. It is a team model, so the mechanism is not driven by moral hazard or agency problems.

4.6 Relation to the literature

- **Versus Adam Smith / Becker–Murphy:** the paper emphasizes that the division of labor is limited not only by market size or coordination costs, but also by the organization's choice of adaptiveness.
- **Versus Bolton–Dewatripont and Garicano:** communication improvements need not increase specialization because they also make local adaptation more valuable.
- **Team theory:** the paper contributes to the Marschak–Radner tradition by endogenizing the division of labor that creates communication and coordination problems.
- **Complementarity literature:** the paper gives a microfoundation for why organizational practices such as broad jobs, employee autonomy, teams, and communication technologies tend to be adopted together.

5 Tables and figures

5.1 Figure 1: task bundling and the importance of coordination

Read:

- The figure plots optimal task bundling t^* against the coordination parameter β.
- There are two cases: high importance of adaptation and low importance of adaptation.
- When adaptation is important, higher β leads to more task bundling.

- When adaptation is less important, t^* is nonmonotone in β: the organization first bundles more tasks, but at high β it returns to specialization.

Main takeaway. Coordination need can produce either broad jobs or rigid specialization. If coordination becomes too important relative to adaptation, the organization may stop adapting and coordinate through strict rules.

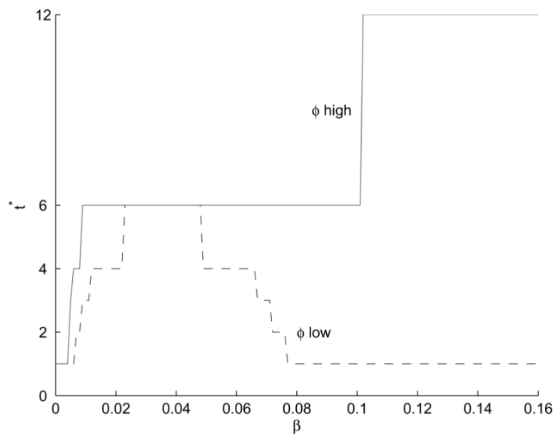

Fig. 1.—Optimal task bundling t^* as a function of the importance of coordination, β, for the case considered in example 1. The continuous line, denoted ϕ high, shows t^* when the importance of adaptation is high, $\phi > \hat{\phi}$, where $\hat{\phi}$ is given in proposition 4; the dashed line, denoted ϕ low, shows t^* as a function of β when the importance of adaptation is low, $\phi < \hat{\phi}$.

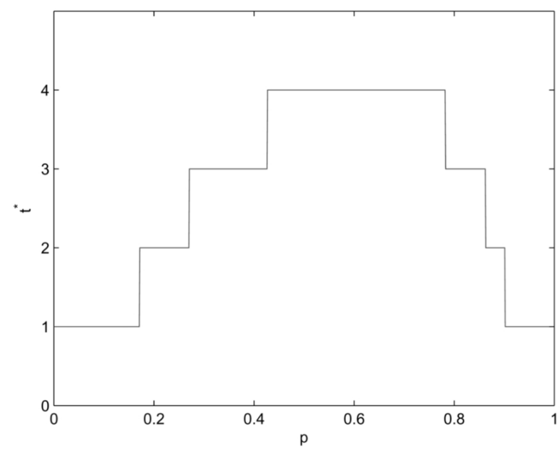

Fig. 2.—Optimal task bundling, t^*, as a function of the quality of communication, p, for the case considered in example 2.

5.2 Figure 2: task bundling and communication quality

Read:

- The figure plots optimal task bundling t^* against communication quality p.
- For low p, better communication leads to more task bundling.
- For high p, better communication eventually leads to more specialization.

Main takeaway. Communication improvements are not always specialization-enhancing. Early improvements can transform a rigid organization into an adaptive one with broader jobs.

5.3 Figure 3: task bundling and the importance of adaptation

Read:

- The figure compares optimal task bundling when communication quality is exogenous with optimal task bundling when communication quality is chosen endogenously.
- As ϕ rises, task bundling rises.
- When communication is endogenous, task bundling responds more sharply because the organization can simultaneously raise communication quality.

Main takeaway. Endogenous communication strengthens complementarities. Organizations can jump from specialized to adaptive designs rather than adjust smoothly.

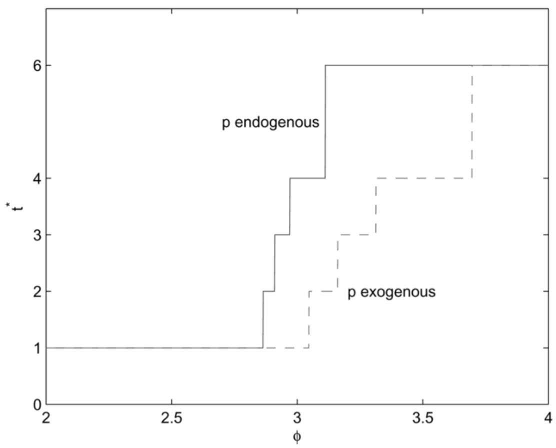

FIG. 3.—Optimal task bundling, t^*, as a function of the importance of adaptation, ϕ, for the case considered in example 3. The thick line, denoted p endogenous, is the level of task bundling when the quality of communication, p^*, is endogenous. The dashed line, denoted as p exogenous, is the level of task bundling when the quality of communication is kept fixed at its optimal level for $\phi = 2$.

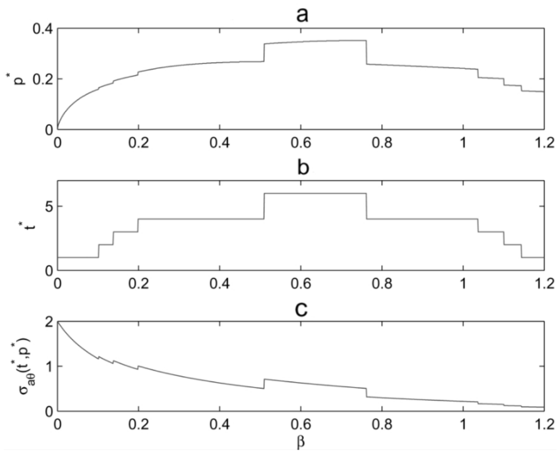

FIG. 4.—Optimal organizational design as a function of the importance of coordination, β, for the case considered in example 3 (continued): a, quality of the horizontal communication, p^*; b, task bundling, t^*; c, employee flexibility, i.e., $\sigma_{a\theta}(t^*, p^*)$.

5.4 Figure 4: endogenous communication, task bundling, and flexibility

Read:

- The figure varies the coordination parameter β with endogenous communication.
- Communication quality p^* first rises and then falls with β.
- Task bundling t^* also first rises and then falls with β.
- Employee flexibility falls within a fixed organizational regime when coordination becomes more important, but it jumps when the organization switches to broader jobs and better communication.

Main takeaway. Communication intensity and broad job design move together. When task interdependence becomes too high, the organization may switch back to a rigid low-communication mode.

5.5 Figure 5: organizational design and communication costs

Read:

- The figure varies the communication-cost parameter δ.
- High δ produces a rigid and specialized organization.
- As δ falls below about 1.205 in the example, the organization jumps from $t^* = 1$ to $t^* = 6$, with $p^* = 1$ and full flexibility.
- Once communication becomes very cheap, below about 0.858 in the example, additional cost reductions again favor more specialization.

Main takeaway. Improvements in communication technology can first reduce specialization and later increase it. The relation is nonmonotone because communication changes the organization's desired level of adaptation.

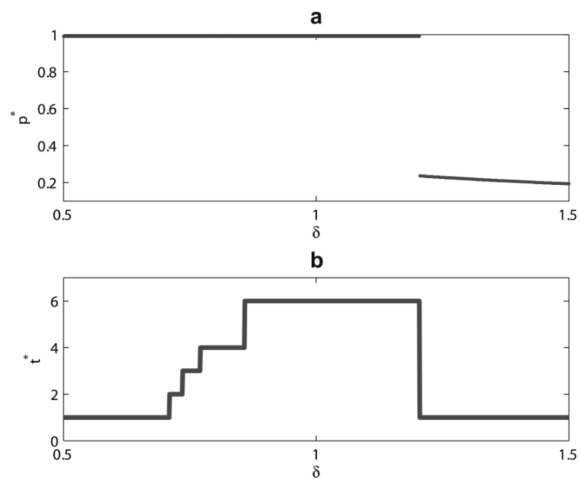

FIG. 5.—Optimal organizational design as a function of the communication cost parameter, δ, for the case considered in example 4: *a*, quality of the horizontal communication, p^*; *b*, task bundling, t^*.

6 Short summary

- **Method.** A team-theoretic model of organizational design with local information, task interdependence, imperfect communication, task bundling, and specialization costs.
- **Core mechanism.** Adaptive organizations use workers' local information, which creates ex post coordination needs. Rigid organizations suppress local adaptation and coordinate ex ante through predictable actions.
- **Main comparative statics.** More uncertainty or more important adaptation leads to broader jobs, more flexibility, and more communication. Stronger returns to specialization lead to narrower jobs and less flexibility.
- **Nonmonotonic results.** More task interdependence or better communication can either increase or reduce specialization, depending on whether the organization responds by bundling tasks or by suppressing flexibility.
- **Main implication.** Modern workplace practices cluster because broad jobs, employee autonomy, and communication technology are complementary.

7 Conclusion

7.1 Core conclusion

The paper argues that the division of labor inside organizations cannot be understood without endogenizing adaptiveness. Organizations choose whether to use local information or suppress it. That choice determines whether coordination is achieved ex ante by rules or ex post through communication and task bundling.

7.2 Economic intuition

- Specialization is efficient when tasks are predictable and coordination can be planned in advance.
- Adaptation is valuable when local circumstances are important and uncertain.
- But adaptation makes actions less predictable, so it increases the need for communication and broader jobs.
- Therefore communication technology can reduce specialization by making adaptive organizations feasible.

7.3 Takeaway

The paper's key lesson is that "better communication means more specialization" is incomplete. Better communication can also make organizations more adaptive. Once workers use more local knowledge, organizations need broader tasks and more horizontal communication to keep actions coordinated. This explains why employee flexibility, teamwork, broad job definitions, and communication technologies often appear together as a cluster of new workplace practices.